



Beaches, dunes, and spits

The rock that is torn from exposed rocky shores by the power of the waves is smashed up and swept along the coast to quieter shores. It settles as beaches of shingle or sand, depending on how sheltered they are. These beaches are constantly reshaped by the waves, creating a variety of distinctive beach types. Meanwhile, the wind can blow beach sand way up beyond the tideline to build high coastal dunes.



POCKET BEACHES

Projecting headlands often enclose small sandy beaches. These form where narrow bands of softer rock between the headlands have been cut back by the waves. The sand often builds up over time, but the headlands stop it from being carried along the coast by the waves, as is the case on more open shores.

▲ COPACABANA BEACH

This world-renowned crescent beach forms part of the seafront of Rio de Janeiro in Brazil.

CRESCENT BAYS

Where headlands of hard rock lie on each side of a wide band of soft rock, the waves create a broad bay with a beach that forms in a sweeping curve. These beautiful crescent beaches are ideal for leisure activities such as swimming and surfing, so they are often big tourist attractions. As a result, many have become famous beach resorts.

LONGSHORE DRIFT

Waves breaking at an angle to the shore throw pebbles and sand along the beach at the same angle. Known as longshore drift, this process can carry beach material away and move it out to sea. Sometimes barriers are placed to slow the process of longshore drift. The sand and pebbles then pile up against these in a zigzag pattern, as seen below.

